

Part B: Test Design Document

Cinema Distribution System

Document type: Test Design (Final Project - Part B)

Group / repository: Enterprise-F-Project / cinema-

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Techniques required: Equivalence Partitioning, BVA, Decision Table, State Transition

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This document derives test cases using formal black-box techniques. For each technique we identify the partitions, boundaries, rules, or transitions, then list the resulting test cases and where they are automated in the repository.

1. Equivalence Partitioning (EP)

1.1 Feature: User registration - email field

RegisterRequest.email is constrained by @NotBlank, @Email, and @Size(max=255). Values fall into mutually exclusive partitions based on email format validity.

Partitions identified

Partition ID	Description	Example	Expected
EP-E1	Valid email format	validation@example.com	Passes Bean Validation
EP-E2	Invalid email format	not-an-email	HTTP 400 Validation failed
EP-E3	Blank / missing email	"" (empty)	HTTP 400 (@NotBlank)

Derived test cases

TC ID	Partition	Input	Expected result	Automation
TC-EP-01	EP-E2	email = "not-an-email"	400; fieldErrors contains email	register_withInvalidEmail_shouldReturnBadRequest
TC-EP-02	EP-E1	validation8@example.com (with valid password)	201 Created (mocked service)	register_withPasswordAtMinimumBoundary_shouldBeAccepted

1.2 Feature: License create - movieId nullability

CreateLicenseRequest.movieId has @NotNull. Partitions: present vs absent/null.

Partition ID		Description	Example	Expected	
EP-M1		Non-null movieId	movieId = 1	Passes @NotNull	
EP-M2		Null movieId	movieId = null	HTTP 400 Validation failed	
TC ID	Partition	Input summary		Expected	Automation
TC-EP-03	EP-M2	start/end dates set; movieId null; clientId 1		400; fieldErrors movieId	createLicense_withNullMovieId_shouldReturnBadRequest

2. Boundary Value Analysis (BVA)

2.1 Feature: Registration password length (@Size(min=8, max=100))

Password length is a numeric range with inclusive minimum 8 and maximum 100. BVA focuses on values just below, on, and (optionally) just above the boundaries.

Boundaries identified

Boundary	Value	Class	Expected
Just below min	length = 7 (e.g. Pass123)	Invalid	HTTP 400
Exact min	length = 8 (e.g. Pass1234)	Valid min	Passes validation
Exact max	length = 100	Valid max	Passes validation (designed)
Just above max	length = 101	Invalid	HTTP 400 (designed)

Derived test cases (implemented)

TC ID	Boundary	Password	Expected	Automation
TC-BVA-01	min-1 (7)	Pass123	400; fieldErrors password	register_withPasswordBelowMinimum_shouldReturnBadRequest
TC-BVA-02	min (8)	Pass1234	201 Created (service mocked)	register_withPasswordAtMinimumBoundary_shouldBeAccepted

2.2 Feature: Movie duration (@NotNull + @Positive)

@Positive requires duration > 0. Therefore 0 is the invalid boundary adjacent to the minimum valid value 1.

Boundary	Value	Class	Expected
Invalid boundary	duration = 0	Invalid	HTTP 400
Min valid	duration = 1	Valid	Passes @Positive
Typical valid	duration = 120	Valid	Passes validation

TC ID	Input	Expected	Automation
TC-BVA-03	duration = 0 (other movie fields valid)	400; fieldErrors duration	createMovie_withZeroDuration_shouldReturnBadRequest

3. Decision table - who may create a license

License creation depends on the caller's role and, for distributors, ownership of the movie. The following decision table combines those conditions.

Conditions and actions

ID	Condition / Action
C1	Caller role is DISTRIBUTOR
C2	Caller role is ADMIN
C3	Movie is owned by the calling distributor
A1	Allow license creation (persist ACTIVE license)
A2	Reject with ForbiddenException

Rules (Y = yes, N = no, - = not applicable)

	R1	R2	R3	R4
C1 DISTRIBUTOR	Y	Y	N	N
C2 ADMIN	N	N	Y	N
C3 Owns movie	Y	N	-	-
A1 Allow	X		X	
A2 Forbidden		X		X

Derived test cases

TC ID	Rule	Scenario	Expected	Automation
TC-DT-01	R1	Distributor licenses own movie	201 / ACTIVE license	LicenseServiceTest.create_success_persist sActiveLicense
TC-DT-02	R2	Distributor licenses another's movie	ForbiddenException	create_whenDistributorLicensesOthersMovie_throwsForbidden
TC-DT-03	R3	Admin licenses any movie	ACTIVE license	create_whenAdmin_canLicenseAnyMovie
TC-DT-04	R4	CLIENT attempts create	ForbiddenException	create_whenClientRole_throwsForbidden

4. State transition testing - rental workflow

A rental moves through statuses. Valid transitions are enforced in RentalService. Invalid transitions must be rejected with BadRequestException.

States

- REQUESTED - initial state after client creates a rental
- ACTIVE - rental in progress; movie marked RENTED
- COMPLETED - rental finished; movie returned to AVAILABLE

Valid transitions

From	To	Notes
REQUESTED	ACTIVE	Mark active; movie availability -> RENTED
REQUESTED	COMPLETED	Direct complete allowed by service rules
ACTIVE	COMPLETED	Mark completed; movie availability -> AVAILABLE

Invalid transitions (examples)

From	To	Expected
ACTIVE	REQUESTED	BadRequestException - invalid transition
COMPLETED	ACTIVE	BadRequestException - invalid transition
REQUESTED	REQUESTED	BadRequestException - invalid transition

State diagram (textual): [REQUESTED] --> [ACTIVE] --> [COMPLETED] ; also [REQUESTED] --> [COMPLETED]

Derived test cases

TC ID	Transition	Expected	Automation
TC-ST-01	REQUESTED -> ACTIVE	Status ACTIVE; movie RENTED	RentalServiceTest.updateStatus_requestedToActive_setsMovieRented
TC-ST-02	REQUESTED -> COMPLETED	Status COMPLETED; movie AVAILABLE	updateStatus_requestedToCompleted_setsMovieAvailable
TC-ST-03	ACTIVE -> COMPLETED	Status COMPLETED; movie AVAILABLE	updateStatus_activeToCompleted_setsMovieAvailable
TC-ST-04	ACTIVE -> REQUESTED	BadRequestException	updateStatus_activeToRequested_throwsBadRequest
TC-ST-05	COMPLETED -> ACTIVE	BadRequestException	updateStatus_completedToActive_throwsBadRequest
TC-ST-06	REQUESTED -> REQUESTED	BadRequestException	updateStatus_requestedToRequested_throwsBadRequest
TC-ST-07	E2E full lifecycle	UI shows Requested, Active, Completed	CinemaSystemTest.rentalLifecycle_license_request_activate_complete

5. Traceability summary

Technique	Feature under design	Key automated evidence
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Technique	Feature under design	Key automated evidence
Equivalence Partitioning	Email validity; null movieId	Phase 6 controller validation tests
Boundary Value Analysis	Password length; duration > 0	Phase 6 Auth/Movie validation tests
Decision Table	License create authorization	LicenseServiceTest create scenarios
State Transition	Rental status workflow	RentalServiceTest + Selenium lifecycle

End of Part B - Test Design Document.